

| Pin Nb | PINs                      | FUNCTIONs        | LABELs         | NOTE                                    |
|--------|---------------------------|------------------|----------------|-----------------------------------------|
| A2     | PE0                       | GPIO_EXTI0       | SYSMCU_IRQ     |                                         |
| A3     | PF3                       | GPIO_EXTI3       | DIB3_IRQ       |                                         |
| A4     | PF2                       | GPIO_EXTI2       | DIB2_IRQ       |                                         |
| A5     | PG3                       | GPIO_Output      | WDG_OUT        |                                         |
| A7     | PM12*                     | USB_OTG_FS_DM    |                |                                         |
| A9     | PM6                       | USB_OTG_HS_DP    |                |                                         |
| A10    | PM3                       | GPIO_Output      | DIB3_CSA       |                                         |
| A12    | PD3                       | LTDC_B1          |                | 24-bit RGB data                         |
| A13    | PE14                      | SPI4_MOSI        | DIB2/DIB5_MOSI |                                         |
| A14    | PD2                       | SDMMC1_CMD       |                | SDcard control                          |
| B1     | PB8*                      | SDMMC1_D4        |                | Not used                                |
| B2     | PE2                       | SPI4_SCK         | DIB2/DIB5_SCK  |                                         |
| B3     | PB5                       | LTDC_R2          |                | 24-bit RGB data                         |
| B4     | PF4                       | GPIO_EXTI4       | DIB4_IRQ       |                                         |
| B5     | PF0                       | I2C2_SDA         | SDA            |                                         |
| B6     | PD7                       | ETH_RMII_REF_CLK |                | Ethernet MII                            |
| B8     | PM9*                      | USB_OTG_HS_ID    |                |                                         |
| B9     | PM5                       | USB_OTG_HS_DM    |                |                                         |
| B10    | PM2                       | GPIO_Input       | DIN2           |                                         |
| B11    | PG0                       | LTDC_R7          |                | 24-bit RGB data                         |
| B12    | PE13                      | SPI4_MISO        | DIB2/DIB5_MISO |                                         |
| B13    | PD1                       | UART4_TX         | RS485_TX/DOUT3 |                                         |
| B14    | PC10                      | SDMMC1_D2        |                | SDcard data                             |
| B15    | PC11                      | SDMMC1_D3        |                | SDcard data                             |
| C2     | PB9*                      | SDMMC1_D5        |                | Not used                                |
| C4     | PB3(JTDO/TRACESWO)        | LTDC_R4          |                | 24-bit RGB data                         |
| C5     | PF1                       | I2C2_SCL         | SCL            |                                         |
| C6     | PG2                       | LTDC_HSYNC       |                | 24-bit RGB control                      |
| C7     | PM13                      | SPI5_MOSI        | DIB4_MOSI      |                                         |
| C8     | PM14                      | SPI5_MISO        | DIB4_MISO      |                                         |
| C9     | PM8*                      | USB_OTG_HS_VBUS  |                |                                         |
| C12    | PE11                      | LTDC_VSYNC       |                | 24-bit RGB control                      |
| C13    | PD0                       | UART4_RX         | RS485_RX/DIN3  |                                         |
| C14    | PA12                      | SPI2_SCK         | DIB1/DIB3_SCK  |                                         |
| C15    | PA11                      | LTDC_B3          |                | 24-bit RGB data                         |
| D1     | PC15-OSC32_OUT(OSC32_OUT) | GPIO_Output      | ETH_RESET      | Ethernet PHY dedicated RESET (optional) |
| D2     | PC14-OSC32_IN(OSC32_IN)   | RCC_OSC32_IN     |                | RTC input                               |
| D4     | PB7                       | ETH_RMII_TXD1    |                | Ethernet MII                            |
| D5     | PB4(NJTRST)               | SPI2_NSS         | DIB1_CSA       |                                         |
| D7     | PD5                       | GPIO_EXTI5       | DIB5_IRQ       |                                         |
| D10    | PM0                       | GPIO_Input       | SD_DETECT      | SDcard detect                           |
| D11    | PE15                      | GPIO_Output      | LED_PWR        |                                         |
| D12    | PC12                      | SDMMC1_CK        |                | SDcard control                          |
| D13    | PA14(JTCK/SWCLK)          | DEBUG_JTCK-SWCLK | SWCLK          | JTAG clock                              |
| D14    | PA9                       | LTDC_B5          |                | 24-bit RGB data                         |
| D15    | PC8                       | SDMMC1_D0        |                | SDcard data                             |
| E1     | PE6                       | GPIO_Output      | DIB5_CSA       |                                         |
| E2     | PE4                       | SPI4_NSS         | DIB2_CSA       |                                         |
| E5     | PB6                       | CEC              | DVID_CEC       | HDMI CEC                                |
| E7     | PD6                       | ETH_CLK          |                | Ethernet MII                            |
| E10    | PG1                       | LTDC_R6          |                | 24-bit RGB data                         |
| E11    | PE12                      | GPIO_EXTI12      | DVID_HPD       | HDMI hotplug detection                  |
| E12    | PA13(JTMS/SWDIO)          | DEBUG_JTMS-SWDIO | SWDIO          | JTAG data                               |
| E13    | PA10                      | LTDC_B4          |                | 24-bit RGB data                         |
| E14    | PC9                       | SDMMC1_D1        |                | SDcard data                             |
| E15    | PC6*                      | SDMMC1_D6        |                | Not used                                |
| F1     | PG14                      | GPIO_Output      | SYSMCU_CS      |                                         |
| F2     | PG12                      | LTDC_G1          |                | 24-bit RGB data                         |
| F3     | PG11                      | ETH_RMII_TX_EN   |                | Ethernet MII                            |
| F12    | PA8                       | LTDC_B6          |                | 24-bit RGB data                         |
| F13    | PC7*                      | SDMMC1_D7        |                | Not used                                |
| F14    | PN1                       | XSPIM_P2_NCS1    |                | FLASH memory control                    |
| G5     | PG13                      | ETH_RMII_TXD0    |                | Ethernet MII                            |
| G11    | PA15(JTDI)                | LTDC_R5          |                | 24-bit RGB data                         |
| G13    | PN3                       | XSPIM_P2_IO1     |                | FLASH memory data                       |
| G14    | PN0                       | XSPIM_P2_DQS0    |                | FLASH memory control                    |

|     |                  |                 |                   |                     |
|-----|------------------|-----------------|-------------------|---------------------|
| G15 | PN11             | XSPIM_P2_IO7    |                   | FLASH memory data   |
| H4  | PF7              | UART7_TX        | UART_TX/DOUT1     |                     |
| H5  | PF6              | GPIO_Output     | DIB4_CSA          |                     |
| H13 | PN10             | XSPIM_P2_IO6    |                   | FLASH memory data   |
| H14 | PN9              | XSPIM_P2_IO5    |                   | FLASH memory data   |
| H15 | PN2              | XSPIM_P2_IO0    |                   | FLASH memory data   |
| J2  | PF9              | LTDC_R0         |                   | 24-bit RGB data     |
| J3  | PF10             | LTDC_R1         |                   | 24-bit RGB data     |
| J14 | PN6              | XSPIM_P2_CLK    |                   |                     |
| K1  | PH0-OSC_IN(PH0)  | RCC_OSC_IN      |                   | RCC XTAL input      |
| K2  | PH1-OSC_OUT(PH1) | RCC_OSC_OUT     |                   | RCC XTAL output     |
| K13 | PN8              | XSPIM_P2_IO4    |                   | FLASH memory data   |
| K14 | PN4              | XSPIM_P2_IO2    |                   | FLASH memory data   |
| K15 | PN5              | XSPIM_P2_IO3    |                   | FLASH memory data   |
| L1  | PC1              | ETH_MDC         |                   | Ethernet MII        |
| L2  | PC2              | SPI2_MISO       | DIB1/DIB3_MISO    |                     |
| L3  | PC3              | SPI2_MOSI       | DIB1/DIB3_MOSI    |                     |
| L4  | PA2              | ETH_MDIO        |                   | Ethernet MII        |
| L5  | PC4              | ETH_RMII_RXD0   |                   | Ethernet MII        |
| L12 | PB12             | LTDC_G5         |                   | 24-bit RGB data     |
| L13 | PD15             | GPIO_Input      | CM5_ID2           |                     |
| L14 | PD14             | GPIO_Output     | USB_FS_SEL        | USB mux selection   |
| M1  | PA0              | LTDC_G3         |                   | 24-bit RGB data     |
| M2  | PA1              | LTDC_G2         |                   | 24-bit RGB data     |
| M3  | PA4              | LTDC_R3         |                   | 24-bit RGB data     |
| M4  | PA7              | ETH_RMII_CRS_DV |                   | Ethernet MII        |
| M5  | PF11             | LTDC_B0         |                   | 24-bit RGB data     |
| M6  | PE7              | UART7_RX        | UART_RX/DIN1      |                     |
| M11 | PO3              | XSPIM_P1_DQS1   |                   | PSRAM control       |
| M12 | PP10             | XSPIM_P1_IO10   |                   | PSRAM data          |
| M13 | PB14             | UART4_DE        | RS485_TX_EN/DOUT2 |                     |
| M14 | PB15             | LTDC_G7         |                   | 24-bit RGB data     |
| N1  | PA3              | LTDC_DE         |                   | 24-bit RGB control  |
| N2  | PA5              | LTDC_CLK        |                   | 24-bit RGB control  |
| N3  | PC5              | ETH_RMII_RXD1   |                   | Ethernet MII        |
| N4  | PB2              | LTDC_B2         |                   | 24-bit RGB data     |
| N5  | PF15             | SPI5_SCK        | DIB4_SCK          |                     |
| N6  | PP12             | XSPIM_P1_IO12   |                   | PSRAM data          |
| N7  | PP14             | XSPIM_P1_IO14   |                   | PSRAM data          |
| N9  | PP2              | XSPIM_P1_IO2    |                   | PSRAM data          |
| N10 | PP5              | XSPIM_P1_IO5    |                   | PSRAM data          |
| N11 | PO2              | XSPIM_P1_DQS0   |                   | PSRAM control       |
| N12 | PP1              | XSPIM_P1_IO1    |                   | PSRAM data          |
| N13 | PD12             | GPIO_Input      | CM5_ID0           |                     |
| N14 | PB11             | LTDC_G6         |                   | 24-bit RGB data     |
| N15 | PB13             | LTDC_G4         |                   | 24-bit RGB data     |
| P1  | PA6              | LTDC_B7         |                   | 24-bit RGB data     |
| P2  | PB0              | GPIO_Output     | VMUX_EN           | Video mux enable    |
| P3  | PF13             | GPIO_Output     | VMUX_SEL          | Video mux selection |
| P4  | PF14             | LTDC_G0         |                   | 24-bit RGB data     |
| P6  | PP11             | XSPIM_P1_IO11   |                   | PSRAM data          |
| P8  | PP15             | XSPIM_P1_IO15   |                   | PSRAM data          |
| P9  | PP3              | XSPIM_P1_IO3    |                   | PSRAM data          |
| P11 | PP0              | XSPIM_P1_IO0    |                   | PSRAM data          |
| P12 | PP7              | XSPIM_P1_IO7    |                   | PSRAM data          |
| P13 | PP8              | XSPIM_P1_IO8    |                   | PSRAM data          |
| P14 | PD13             | GPIO_Input      | CM5_ID1           |                     |
| P15 | PB10*            | ETH_MII_RX_ER   |                   | Ethernet MII        |
| R2  | PB1              | GPIO_EXTI1      | DIB1_IRQ          |                     |
| R4  | PE8              | GPIO_EXTI8      | TFT_IRQ           |                     |
| R5  | PE10             | GPIO_Output     | LED_nACT          | Active LED          |
| R7  | PP13             | XSPIM_P1_IO13   |                   | PSRAM data          |
| R8  | PP4              | XSPIM_P1_IO4    |                   | PSRAM data          |
| R10 | PO4              | XSPIM_P1_CLK    |                   | PSRAM control       |
| R11 | PP6              | XSPIM_P1_IO6    |                   | PSRAM data          |
| R13 | PO0              | XSPIM_P1_NCS1   |                   | PSRAM control       |